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The efficacy of Vespex® wasp bait to control *Vespula* species (Hymenoptera: Vespidae) in New Zealand

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ABSTRACT

Transitioning from trials to pest control tool, the efficacy of controlling wasps *Vespula vulgaris* and *V. germanica* using Vespex® wasp bait was tested in 2015. Vespex® is a protein-based bait matrix with 0.1% fipronil. Five sites ranging <300 to >2000 ha and over 5500 ha in total had bait stations placed 300 × 50 m apart and received one application of Vespex®. Wasp nest traffic rates declined 93% in four days at one site, and over 97% after 20–38 days at all other sites. Measured honeydew availability also improved post wasp control. Further demonstrating feasibility, five different agencies worked closely with government leading many programme aspects. Vespex® was made more widely available in New Zealand towards the end of 2015. Web page information showed ~30% of all territorial authorities nationally are engaging in pest wasp problems. This signals a future potential in the way that pest wasps might be managed in a community context for social, economic and biodiversity conservation objectives.

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Introduction

For New Zealand, the significant negative ecological consequences of two invasive wasp species *Vespula vulgaris* (common wasp) and *V. germanica* (German wasp) are widely reported (Beggs 2001; Beggs et al. 2011; Lester et al. 2013; Ward 2013). Both wasp species have colonised lowland to montane areas throughout (Clapperton et al. 1994) and *V. vulgaris* nests are also occasionally reported above timber-line (Beggs 1991; Clapperton et al. 1994). *V. germanica* established in the 1940s and spread across much of New Zealand and then subsequently declined in some areas after the spread of *V. vulgaris* during the 1980s (Clapperton et al. 1994). *V. vulgaris* became dominant over *V. germanica* in upper South Island areas of beech forest (*Fuscospora* species) where indigenous honeydew scale insects (*Ultracoelostoma* species: Coelostomidiidae) occur on the trunks and main branches (Harris et al. 1991b; Beggs et al. 2005). *V. germanica* remains prevalent in many rural areas and in mild climates at low altitudes near the coast (Clapperton et al. 1994).

High summer densities of both wasps occur wherever there is an abundant carbohydrate resource such as insect honeydew (Beggs 2001; Gardner-Gee & Beggs 2013). In some rural and peri-urban regions, nest or colony densities of *V. vulgaris* and *V. germanica* together range from <1 per hectare (Moller et al. 1991a; Harris and Oliver 1993) to >12 nests per hectare where a honeydew food source is present (Moller et al. 1990; Barlow et al. 2002; Beggs et al. 2008). Harris

and Oliver (1993) discuss wasp density in shrubland and pasture and noted a patchy distribution of wasp nests. Human contact with wasps occurs in towns and cities as well as more remote areas (Moller et al. 1991a; Sheppard & Urquhart 1991; Clapperton et al. 1994).

Local authorities including Regional Councils provide advice on pest wasp control and most urban pest control companies include wasp nest control among their services. McIntyre and Helstrom (2015) calculate health-related costs of *Vespula* species wasps to be ~\$1 M (New Zealand currency) per annum in New Zealand. This includes doctor's visits, death from severe anaphylaxis reaction to venom and vehicle accidents (MacIntyre & Helstrom 2015). An overall estimate of economic costs of wasps to NZ is ~\$130 M per annum and this is largely associated with farming, but also includes almost \$9 M in direct costs to apiarists (MacIntyre & Helstrom 2015). Limited quantitative data are available to estimate the effect on social well-being and people's enjoyment of outdoor activities in New Zealand, but this is also likely to be significant (MacIntyre & Helstrom 2015, and see Kerr & Sharpe 2008). A brief survey of leaders in outdoor recreation carried out by MacIntyre and Helstrom (2015) demonstrated *Vespula* species annoyed people involved in tourism and recreation and that carrying out a robust survey of this kind is likely to record a significant economic and social cost. Sheppard and Urquhart (1991) surveyed 1000 New Zealanders in 1991 for their views

on the seriousness of pests. They note “Nearly a third of respondents indicated that they had a wasp problem where they lived with rural, town and city areas being equally represented” (Sheppard & Urquhart 1991).

Until 2015, the only method of control publicly available for *Vespula* wasps has been to find individual nests. In many situations, this is not practical where a site has many nests and time spent locating and destroying them is not justified, or a site is larger than 1–2 hectares in size. Attempting to lure and catch worker wasps generally has little effect on nest viability since queens would remain unharmed (Ward 2013). To effectively control wasps by trapping, it may be more effective to develop a method that attracts queens before they establish a nest rather than targeting worker wasps (Ward 2013). Trials in New Zealand have shown that the use of fresh protein bait containing 1 g/kg fipronil-active ingredient in elevated weatherproof bait stations over 48 hours is sufficient to control wasp colonies (Harris & Etheridge 2001; Harper et al. 2016). Using this method, it is not necessary to find wasp nests since worker wasps carry bait back to share in the nest. Wasp activity can be reduced by more than 90% (Harris & Etheridge 2001; Long et al. 2014).

Areas as large as 500 hectares in size have been treated with wasp bait stations placed on trees at 50 m intervals along lines that are spaced 300 m apart (Long et al. 2014). Such effective reductions among *Vespula* species wasp populations are also reported elsewhere including Tasmania (*V. vulgaris* together with *V. germanica*, Bashford 2010), Patagonia (*V. germanica*, Sachmann et al. 2001) and Hawai'i (*V. pensylvanica*, Hanna et al. 2011). Harper et al. (2016) showed an 80% reduction in wasp nest activity can be expected within 100 metres of a single line or a single cluster of bait stations in an area with wasp nest density greater than five nests per hectare. The New Zealand Environmental Protection Authority as the regulatory agency

has approved a wasp baiting method and bait (ERMA 2006) marketed as Vespex® for use in New Zealand. This article explores the transition from scientific trials to an available pest control tool. We test the operational efficacy of wide-scale control and community agency participation in control for both common wasp *V. vulgaris* and German wasp *V. germanica* using Vespex® wasp bait in 2015. Specifically, managing wasps for ecosystem protection was repeated at five sites.

Materials and methods

Sites and establishing bait station networks

Efficacy of wasp baiting was tested at five different sites managed by the same agency (New Zealand Department of Conservation). However, staff and partner agencies carrying out the work were unique to each site. Vegetation and climate, summarised in Table 1, are different at each site. However, all have a component of *Fuscospora* species (Nothofagaceae) beech forest. Endemic scale insects (*Ultracoelostoma* species: Coelostomidiidae) on the trunks and main branches of *Fuscospora* species trees produce honeydew as a carbohydrate food resource. *Vespula vulgaris* was the primary wasp species at all sites but *V. germanica* was also present. These species have largely overlapping diet and are opportunistic towards protein resources (Harris et al. 1991a; Harris & Oliver 1993). Therefore, treatment and monitoring methods are applicable to either wasp species (Moller et al. 1991a) and we did not discriminate between them in our analysis.

Small folded plastic weatherproof bait stations (capable of containing 30 g of paste bait) were placed on trees at 1.5 metres above ground in an array of 50 × 300-metre spacing and a maximum spacing of 50 m apart along all site boundaries. That is, a series of parallel walking lines were marked at 300-metre intervals from each other and a bait station was placed every 50 metres along each line. Department of

Table 1. Description of sites and treatment dates where bait station wasp control was piloted.

Site, region and area (ha)	Treatment and non-treatment site vegetation and context	Treatment days ^a
Rotoiti Nature Recovery Project, Nelson, 1749 ha	Cool montane mixed <i>Fuscospora</i> -dominated beech forest. Elevation: 620–1300 m. Surroundings include extensive similar forest, lake margin, township of St Arnaud and agriculture. Non-treatment site between 1 and 2 km north of the site boundary	Feb 10–17
Craigieburn Basin, Canterbury, 2477 ha	Cool montane <i>Fuscospora</i> and <i>Lophozonia</i> beech forest. Partly fragmented by fire and includes short tussock grassland, riverbed, shrublands and exotic trees. Elevation: 700–1000 m. Surroundings include similar forest, Castle Hill Village, exotic forest and agriculture. Non-treatment site 6 km east in similar vegetation and aspect	Feb 10–22
Falls River Block, Nelson, 736 ha	Cool conifer-broadleaved-beech forest. <i>Fuscospora</i> and extensive secondary shrubland. Elevation: 0–300 m. Surroundings include similar vegetation, marine margin, Torrent Bay Village Campsite and other holiday amenities. Non-treatment site 1 km north in similar vegetation	Feb 11–19
Pelorus Bridge, Marlborough, 217 ha	Tall podocarp and mixed beech forest. Elevation: 30–400 m. Surroundings include similar vegetation, agriculture, exotic forest and Pelorus Bridge Campsite. No non-treatment site reported	Feb 16–23
Bottle Rock Peninsula, Marlborough, 432 ha	<i>Dysoxylum</i> (Kohekohe) hardwood forest and shrub. Extensive secondary shrub and small patches of <i>Fuscospora</i> beech. Elevation: 0–400 m. Surroundings include marine margin and similar vegetation. No non-treatment site reported	Mar 11–17

^aIn February or March 2015.

Conservation staff and a community agency at one site, Nelson Lakes National Park, had a long history of developing and testing wide-scale wasp control (Butler 2003; Long et al. 2014). Staff and partner agencies at the other four sites had less or no experience of deploying bait station control or of measuring the level of success.

Assessing wasp activity on protein foods

Prior to carrying out wasp baiting and to ensure success, an assessment of the level of wasp activity on non-toxic protein bait (Harper et al. 2010) was carried out at all five sites. This is performed under warm day-time conditions (typical maximum 18–23 °C) without rain. Twenty shallow trays containing approximately 15 g of non-toxic protein (fish-based cat-food or canned fish) were placed within the treatment area at 5 m intervals and left for one hour. Then, an instantaneous count of the number of wasps observed on each tray was totalled. A total of at least 20 wasps/20 trays in two or more areas of a site for treatment indicate sufficient activity on protein foods for successful bait station treatment (Harris et al. 1991b; Harper et al. 2010).

At four sites, wasp activity on non-toxic canned fish baits met the threshold minimum 20 wasps per 20 trays and exceeded it by the first week of February 2015 (Table 2). At one site, Bottle Rock Peninsula, this threshold was reached in the last week of February 2015 (Table 2). These results led to decisions in each case to proceed with the use of Vespex® and the dates are noted in Table 1.

Treatment

As soon as the wasp activity threshold on non-toxic bait has been met for a site (generally, late January or in February) and two or more fine weather days (daily maximum temperatures above 17 °C and no rain) were forecast, then each bait station was filled with 20–30 g of Vespex® (a protein wasp bait containing 1 g/kg of fipronil-active ingredient). All residual bait was removed from stations 3–7 days later.

Monitoring for treatment effect on wasps

Monitoring the level of operational success in reducing wasp activity involved measuring traffic rates at nest entrances (Malham et al. 1991). The number of wasps exiting a nest in a 60-second period was counted followed by a count of the number entering the nest. This was repeated between two and five times and then the average total number of wasps exiting plus entering over one minute was calculated. Nine to twelve wasp nests were found and monitored within the bait station network before and after the operation. In two areas (Nelson Lakes National Park and Abel Tasman National Park), another 10 or 13 nests were monitored in an equivalent non-treatment area, a minimum 1 kilometre from treatment (Table 1). A reduction in wasp activity at each site was detected by recording activity counts at a nest approximately one week before and two weeks after bait station control and calculating the difference between counts for each nest. Counts were repeated again more than four weeks later at three sites. Mean percentage differences (and standard error) are reported for nine or more nests at all treatment sites and also for two associated non-treatment sites at Rotoiti Nature Recovery Project and Falls River, respectively.

Monitoring an indicator of ecosystem response

Detecting ecosystem level response involved placing and monitoring permanent quadrats on *Fuscospora* spp. beech trees with honeydew from *Ultracoelostoma* scale insects present (see Beggs & Wardle 2006). Quadrats (5 × 50 cm) were established at breast height on either black beech (*Fuscospora solandri*) or red beech (*F. fusca*) where honeydew scale insects occupied tree stems. In December 2014, 24 quadrats were established within the five wasp treatment sites and at four associated untreated sites, which were a minimum of 1 kilometre from a treatment area (Table 1). The total number of honeydew insect anal tubes with a droplet as wide or wider than the tube was counted in each quadrat. A time series of mean quadrat counts for the different treatment and non-treatment sites was obtained. The timings of these counts were as follows: (1) at establishment in December (prior to wasp numbers

Table 2. Results of assessment for wasp feeding activity on protein.

Pilot site	Date	Lines of trays scored				Mean
		1	2	3	4	
Rotoiti Nature Recovery Project	27 Jan 2015	12.5				12.5
Falls River	22 Jan 2015	1.8	1.3			1.55
Falls River	4 Feb 2015	12.2	4.7	5.2	4.1	6.55
Craigieburn Basin	13 Jan 2015	0.3	0.15			0.23
Craigieburn Basin	22 Jan 2015	1.15	0.15	0.4	0.7	0.6
Craigieburn Basin	29 Jan 2015	1.8	0.7	1.25	1.25	1.25
Pelorus Bridge	12 Feb 2015	9.5	5.5	1.5		5.5
Bottle Rock Peninsula	27 Feb 2015	1.15	2.05			1.6

Each line includes 20 trays. And each score is the total number of wasps counted divided by 20. A score ≥ 1.0 is sufficient activity to proceed with bait station treatment.

reaching high levels), (2) approximately one week prior to wasp treatment, and (3) approximately two weeks after treatment. For each site, the change in honeydew counts from before to after the treatment time was calculated for each quadrat ($n = 24$) and the mean change compared between the treatment and associated non-treatment using 95% confidence intervals and two-sided Welch two-sample t -tests. Mean honeydew counts with confidence intervals are also graphed in time series for each site. At three sites, there was an additional count undertaken 4 weeks after treatment to assess longer term impact of wasp reduction on honeydew levels, but no statistical test was applied.

Reporting the roles and motivation of partner agencies for wasp control

During the 2015 pest wasp season, the Department of Conservation proposed to supervise wasp control piloted on public conservation lands and to accept assistance offered by other agencies (listed in Acknowledgements section). Agency project contributions across seven categories were scored and motivation for either biodiversity conservation or public safety, plus enjoyment of a wasp-free environment, was noted.

Results

Monitoring for treatment effect on wasps

Vespula spp. wasp nests monitored at the five treatment sites all had a reduction in relative activity by at least 93% (Table 3). A 93% reduction in wasp activity was recorded at the Pelorus Bridge site only four days after treatment. In the four other wasp-baiting operations where nest counts were done 20 days to 38 days later, 97%–100% reductions in wasp activity were recorded (Table 3). To ensure that the effect was due to the treatment, wasp nests were also monitored in nearby non-treatment areas in two cases (Rotoiti Nature Recovery Project and Falls River) and these gave confidence around the treatment effect on wasps (Table 3). At Rotoiti Nature Recovery Project, mean wasp traffic rate among untreated nests increased (approximately $60\% \text{ SE} \pm 51\%$) by March 20 (Table 3).

Monitoring beech scale honeydew

Figure 1 and Table 4 show that the number of honeydew droplets in *Fuscospora* tree trunk plots increased at four sites relative to the associated non-treatment

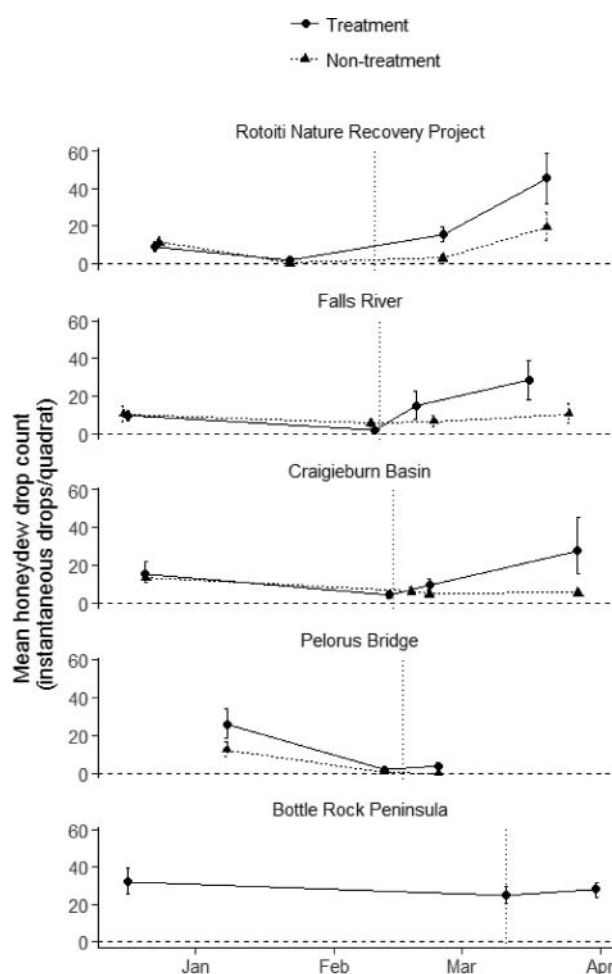


Figure 1. Honeydew monitoring results. Observing response to wasp treatment for each of five sites where wasps are abundant harvesting honeydew and then wasps are treated reducing their harvest. Data are the means (with 95% confidence intervals) from counts of honeydew drops in a 50 cm × 5 cm quadrat on *Fuscospora* beech trees; $n = 24$ quadrats. Circles connected by solid lines for counts in wasp treatment area. Triangles connected by dashed lines for counts in areas not treated for wasps. Wasp treatment date is indicated by vertical dashed line.

Table 3. Wasp activity and changes in wasp activity recorded in treatment areas and in non-treatment areas at five sites.

Site	First count	T/N	Days post	2nd count	Difference	Days post	3rd count	Difference	<i>n</i>
RNRP	41.1	T	16	1.9	−94% ± 3%	38	0.7	−98% ± 2%	12
	46.7	N	16	62.5	+43% ± 33%	38	70.8	+60% ± 51%	10
FR	45.5	T	9	0.5	−98% ± 1%	33	0	−100%	10
	66.6	N	12	60.6	−8% ± 18%				13
CB	60.5	T	9	2.2	−96% ± 1%	37	0	−100%	10
PB	79.2	T	4	5.1	−93% ± 4%				11
BRP	70.6	T	20	2.3	−97% ± 3%				9

RNRP, Rotoiti Nature Recovery Project; FR, Falls River; CB, Craigieburn Basin; PB, Pelorus Bridge; BRP, Bottle Rock Peninsula. Counts are the mean count of wasps both entering and leaving (n) nests in 60 s. This is a metric describing wasp activity in their environment. T/N, T = nests inside treatment site versus N = nests in a non-treatment site. Days post = the number of days, post-treatment date when nest activity counts were recorded. Difference = mean difference in activity counted at nests post-treatment ± standard error. n = number of wasp nests.

Table 4. Testing the significance of honeydew response to treatment versus non-treatment.

Site	Treatment mean difference	Non-treatment mean difference	t-value	df	p	95% CI
RNRP	13.7	2.5	4.892	28.9	0.001	6.47–15.78
FR	12.7	0.9	3.345	24.6	0.003	4.54–19.10
CB	5.5	–1.5	6.930	34.9	0.001	4.98–9.10
PB	2.5	–0.6	3.798	24.5	0.001	1.41–4.76
BRP	3.0	–	–	–	–	–

Welch two-sample *t*-test with unequal variances was used. Treatment and non-treatment mean differences = mean of pairwise differences between quadrat counts before and after wasp treatment date. 95% CI indicates the confidence interval for calculated values of Students *t*. Site names are abbreviated as in Table 3.

sites ($p < 0.005$ for all four sites – see Table 4). The Bottle Rock Peninsula non-treatment monitoring site was damaged which invalidated the data. Data for Rotoiti Nature Recovery Project shows that honeydew droplet counts increased significantly in the non-treatment area when re-measured 15 days after treatment date, but the level of increase was lower than that observed in the associated treatment site (Figure 1). Honeydew droplet counts in the Rotoiti Nature Recovery Project continued to increase 4 weeks after, but where wasps were controlled, the increase appears greater (Figure 1). The summer trend in honeydew availability at all five sites suggests pest wasps impact this resource prior to Vespex® treatment and that this impact continues till at least late March.

Reporting the roles and motivation of partner agencies for wasp control

Table 5 indicates similarities and differences among roles carried out by five different partner agencies that contributed to pest wasp control. All five agencies contributed to track network maintenance and establishment of bait stations on trees. All were involved in field operations and all carried out an advocacy role locally and in media (Table 5). Three out of five showed leadership by identifying the site priority and four out of the five agencies helped fund wasp control (Table 5). Finally, partner agencies were primarily motivated to carry out wasp control for biodiversity conservation. However, public safety and enjoyment of

a wasp-free environment were also important for two agencies (Table 5).

Discussion

Controlling wasps using a single annual application of a protein-based bait containing 1 g/kg fipronil has often shown promise as a management tool (Patagonia–Argentina, Sackmann et al. 2001; Tasmania–Australia, Warren & Statham 2002; Bashford 2010; Hawai'i-USA, Foote et al. 2011; Hanna et al. 2013; New Zealand, Harris & Etheridge 2001; Harper et al. 2016). This study demonstrates feasibility for applications at scales >200 ha (see Harris & Etheridge 2001), and in two cases much greater than 1000 ha (Table 1). Measures of wasp activity at nests showed considerable success. Over 98% reduction was recorded in three cases that were re-measured 33–38 days post-treatment (Table 3). Honeydew from the native scale *Ultra-coelostoma* species on *Fuscospora* species trees is an important resource for ecosystems (Thomas et al. 1990; Beggs 2001; Beggs & Wardle 2006; Dhami et al. 2013). Monitoring showed that wasps deplete this resource over many weeks (see also Moller et al. 1991b) and that while wasp treatment is beneficial (Table 4), wasps have an impact prior to treatment (Figure 1). Treatment date is determined by testing wasp activity on protein foods, and treating as early as possible is likely to provide the maximum benefit.

Another aspect of feasibility was shown by the close engagement of five different partnering agencies working with the government (Table 5). This demonstrates that the pest wasp issue is widely recognised beyond government agencies and that a successful programme of wide-scale wasp control could be supported by communities. People involved want to enjoy an environment free of pest wasps; however, the primary motivation on public conservation lands and parks remained protection of biodiversity from wasps (Table 5).

The government agency responsible for most protected natural areas in New Zealand (Department of Conservation) has trialled and can widely adopt annual bait station wasp control. However, the bait marketed as Vespex® was made more widely available in New Zealand towards the end of 2015. Regional Councils have a primary responsibility for regulation and advice

Table 5. Identifying the contribution of partner agencies to wasp control at five sites.

Partner contribution	Coded sites				
	RNRP	FR	CB	PB	BRP
Identified the site priority		✓	✓	✓	
Planning and logistics	✓		✓		✓
Site maintenance	✓	✓	✓	✓	✓
Field operational support	✓	✓	✓	✓	✓
Conservation advocacy	✓	✓	✓	✓	✓
Funder		✓			✓
Fund raiser			✓	✓	
Partner motivation^a					
Biodiversity conservation	1	1	2	1	1
Public safety and enjoyment of a wasp-free environment		2	1		

Abbreviated sites: RNRP, Rotoiti Nature Recovery Project; FR, Falls River; CB, Craigieburn Basin; PB, Pelorus Bridge; BRP, Bottle Rock Peninsula.

^a 1 = primary motivation and 2 = secondary motivation.

for pest control. We note over 50% of Regional Councils provide advice on their web pages to manage public complaints and enquiries concerning wasp control. A similar number of Councils have registered staff as approved users for Vespex®. Web page information (September 2016) showed ~30% of all territorial authorities nationally are engaging in pest wasp problems. This signals a future change in the way that pest wasps might be managed for social, economic and biodiversity conservation objectives.

Potentially taking this a step further, the New Zealand government in July 2016 stated an ambitious goal for the country to be predator-free by the year 2050 (Norton et al. 2016). Key to this idea was more coordination among community conservation organisations for pest animal control (Norton et al. 2016) and, in the near future, this could include pest wasps.

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